NOVA SCHOOL OF SCIENCE & TECHNOLOGY

Interação Pessoa-Máquina

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Individual Assignment

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Good User Interface Design: Capsule Coffee Machine

Objective
Allow users to quickly and easily brew a specific size of coffee using a single button push.

Good aspects:

- **High Learnability:** Simple layout with usually only two intuitive buttons (e.g., Short Espresso vs. Long Coffee).
- **Clear Feedback:** Status LEDs blink to indicate heating up, stay solid when ready to brew, and blink rapidly if there is an error or water is missing.
- **Physical Constraints:** The capsule slot is shaped specifically so the capsule only fits in the correct orientation.
- **Consistency:** Operation remains identical across different models, reducing cognitive load for new users.

Bad aspects:

- Limited control over water temperature or pressure for advanced users.
- Lack of explicit visual feedback on water level if the reservoir is located at the back.





Explanation:
I consider this a good user interface design because it minimizes cognitive effort and provides immediate visual feedback. The physical constraints prevent user errors during capsule insertion, and the overall system image aligns perfectly with the user's conceptual model of making coffee quickly.

Sugestions

Add a front-facing transparent water indicator or a low-water LED indicator to prevent the pump from running dry when the user forgets to check the rear reservoir.

Bad User Interface Design: Row of Identical Light Switches

Objective
Allow users to independently turn on and off different sets of lights in a room using a panel of wall switches.

Good aspects:

- Low production and installation cost using standardized, compact hardware.
- Simple physical interaction (flipping a switch up or down).

Bad aspects:

- **Arbitrary Mapping:** The linear layout of the switches does not match the spatial distribution of the light fixtures in the room.
- **Frequent Description Errors:** Users frequently press the wrong switch because all switches look and feel identical and are placed right next to each other.
- **Missing Feedback/Signifiers:** There are no labels or visual signifiers indicating which switch controls which light fixture.
- **High Trial-and-Error:** First-time users are forced to flip switches randomly until they find the correct one.



Explanation:

I consider this a bad user interface design because it completely fails at natural mapping. It forces the user to rely entirely on memory (*knowledge in the head*) rather than placing information in the interface (*knowledge in the world*). This design mismatch directly leads to frequent execution slips and description errors.

Suggestions for Improvement:

1. **Apply Natural Mapping:** Arrange the switches in a grid layout that mirrors the actual physical layout of the lights in the room (e.g., top-left switch controls the top-left light).
2. **Add Signifiers:** Include clear icons or labels above each switch.
3. **Tactile/Visual Differences:** Use different shapes or small status LEDs so users can distinguish switches by touch or see their state in the dark.